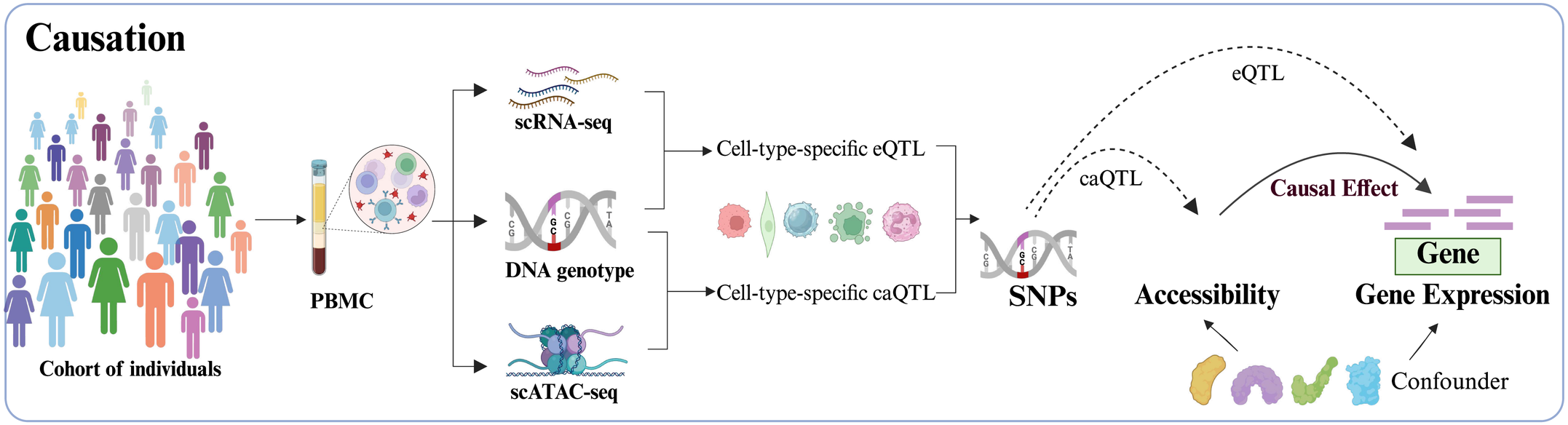
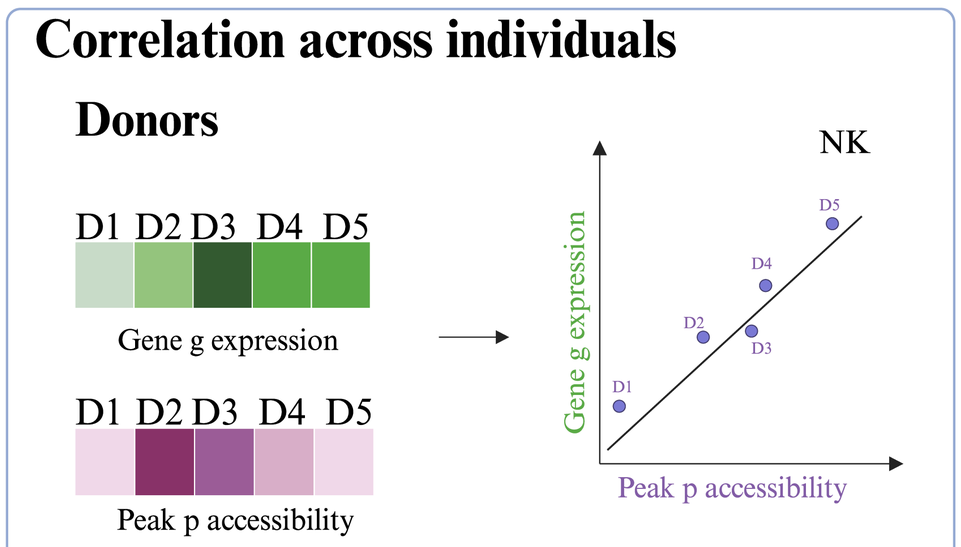
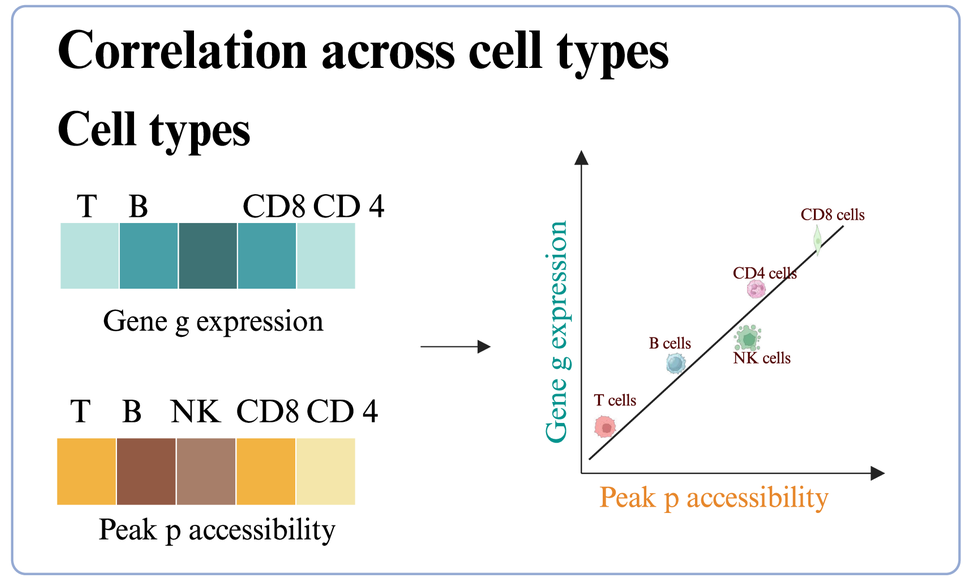
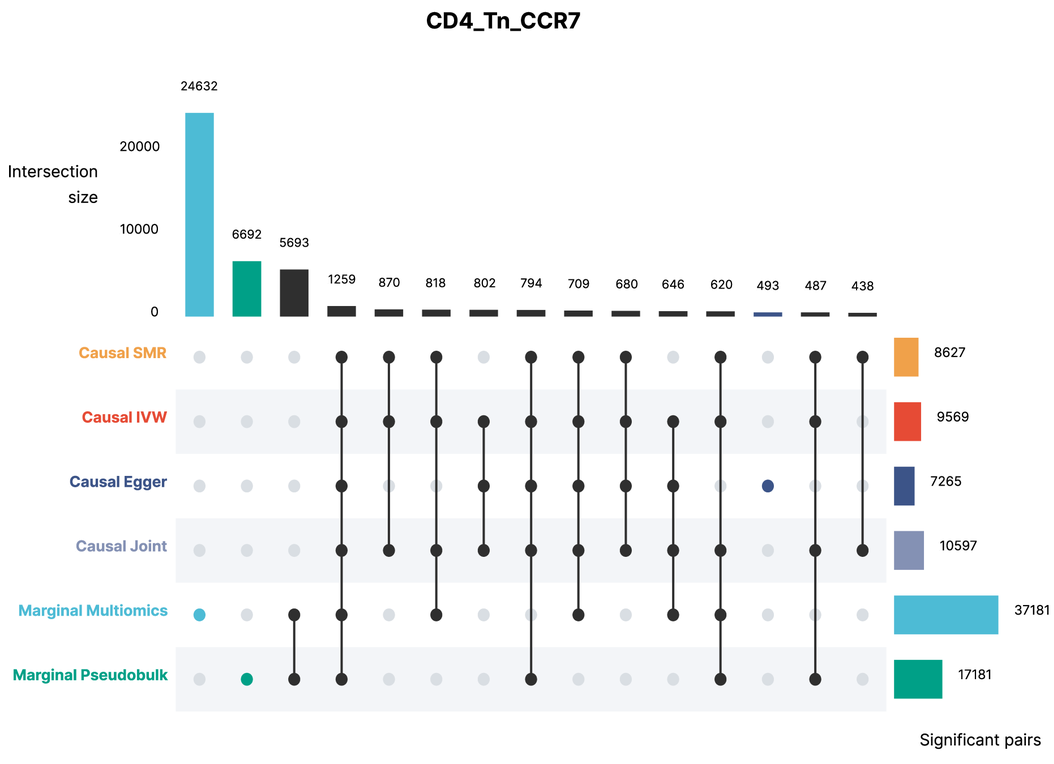
A


Causation
Cohort of individuals
PBMC
scRNA-seq
DNA genotype
scATAC-seq
Cell-type-specific eQTL
Cell-type-specific caQTL
SNPs
eQTL
caQTL
Causal Effect
Gene
Accessibility
Gene Expression
Confounder

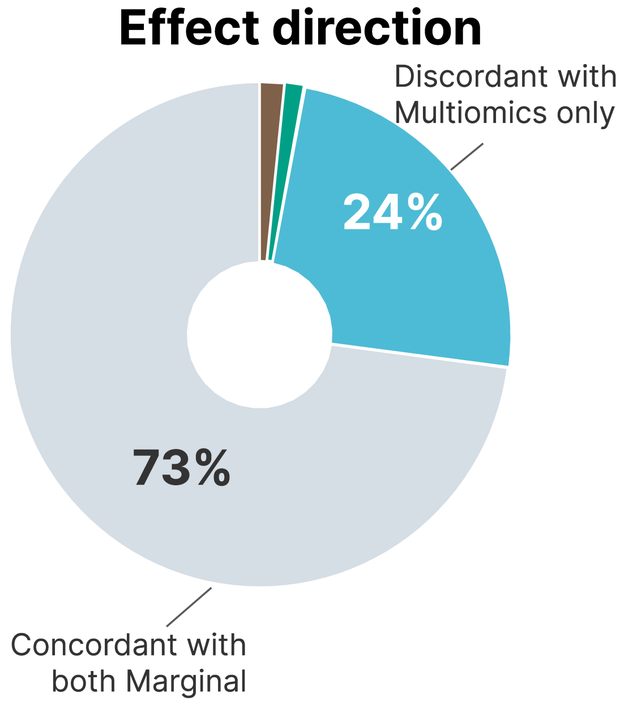

Correlation across individuals
Donors
D1 D2 D3 D4 D5
Gene g expression
D1 D2 D3 D4 D5
Peak p accessibility
NK
Gene g expression
Peak p accessibility


Correlation across cell types
Cell types
T B CD8 CD 4
Gene g expression
T B NK CD8 CD 4
Peak p accessibility
CD8 cells
CD4 cells
B cells
NK cells
T cells
Gene g expression
Peak p accessibility

B


CD4_Tn_CCR7
Intersection size
24632
6692
5693
1259
870
818
802
794
709
680
646
620
493
487
438
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
8627
9569
7265
10597
37181
17181
Significant pairs

C


Effect direction
Discordant with Multiomics only
24%
73%
Concordant with both Marginal